DIRECTORY

The spread of rediscovered classical and Islamic texts throughout the 1500s incited an intellectual revolution across Europe. Defiant thinkers, championing observation and testable theories, modernized the scientific mindset with their new understanding of knowledge based on empiricism.

JACOPO BERENGARIO DA CARPI

1460–1530

Italian physician Jacopo Berengario da Carpi was one of the first to use drawings from nature to illustrate an anatomy book and the first to describe heart valves. The son of a surgeon, he was fascinated by human anatomy and claimed to have dissected several hundred bodies. Berengario studied medicine in Bologna, and went on to teach at the university there. He gained a reputation across Italy for his skill and declined a request to become the pope's doctor. He died a wealthy man.

PARACELSUS

1493–1541

Philippus Aureolus Theophrastus Bombastus von Hohenheim, who called himself Paracelsus (meaning "above or beyond Celsus," a Roman scholar), was a Swiss physician and alchemist who established the role of chemistry in medical practice. Inspired by the folk medicine he learned about on his travels, Paracelsus advocated studying nature and using experiments to learn about the human body. He believed that disease was caused by external influences and that it could be treated with external substances, given in the correct dose. He thought that a goiter was caused by minerals that are found in drinking water and introduced the use of mercury as a cure for syphilis (now obsolete). Most modern vaccines stem from Paracelsus's belief that that which makes a man ill can also cure him.

MICHAEL SERVETUS

1509–1553

Spanish theologian and physician Michael Servetus discovered pulmonary circulation (that aeration of the blood occurs in the lungs and not, as Claudius Galen thought, in the heart) almost incidentally as part of his theological quest to understand the soul, which he claimed could be found in the blood. Both Catholics and Protestants deemed his religious views heretical, and he was burned at the stake in Geneva, Switzerland, in 1553, along with almost every copy of his book.

ANDREAS VESALIUS

1514–1564

Flemish-born doctor Andreas Vesalius studied medicine at the University of Padua, Italy, and became professor of surgery there aged 23. Anatomists of the time relied not on dissection, but on the work of Greek and Roman scholars such as Claudius Galen, whose knowledge of anatomy rested on examining animals. Vesalius challenged this tradition, dissecting human bodies in front of his students and creating an accurate record of what he saw in his 1543 masterpiece *De Humani Corporis Fabrica* (*On the Fabric of the Human Body*). He was physician to the Holy Roman Emperor Charles V, then to Philip II. He died in obscurity on a Greek island following a shipwreck.

GABRIELE FALLOPPIO

1523–1562

Due to lack of funds, Gabriele Falloppio initially joined the clergy, but when his situation improved, he went to study medicine instead. The Italian's skill at dissection earned him the position of professor of anatomy at Pisa in 1548, and he took over from Andreas Vesalius at the University of Padua 3 years later. Falloppio is best known for his study of human reproductive organs and for describing the narrow tubes that connect the ovaries to the uterus (fallopian tubes). He also studied syphilis and designed the first condom as a means of preventing spread of the disease.

WILLIAM GILBERT

1544–1603

After years of painstaking experiments, William Gilbert concluded that the Earth was magnetic. The English physician